



9.4.5 Wrap-Up: Cool Down

The last three lessons have involved writing quadratic functions to represent the relationship between two quantities from a graph, a table, and a written description.

1. Create a sales pitch to convince your classmates which representation you feel is the easiest for writing quadratic functions.

Unit 9, Lesson 5: Solving Quadratics Using Square Roots



Benchmarks

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can solve a quadratic by taking the square root.
- ☐ I can determine if a solution to a quadratic is non-real.
- ☐ I can interpret solutions to quadratics given in a real-world context as viable or nonviable.



9.5.1 Warm-Up: Floor Tiles

Aniyah is renovating her bathroom. She has chosen the floor tile shown. The design uses 4 tiles. The 4 tiles have an area of 2.56 square feet (ft^2). Aniyah wants to determine the measurements, in feet, for each tile.



1. Determine the measurements, in feet, for each tile.
2. Discuss with your partner how the equation $4x^2 = 2.56$ represents the context. Summarize the discussion.



9.5.2 Exploration: Solving Quadratics Using Square Roots

1. The steps for solving $4x^2 = 2.56$ are shown. Complete the table by completing the explanation for the tile context.

Step	Tile Context
$4x^2 = 2.56$	Four square tiles with area x^2 have an area of 2.56 ft^2
$\frac{4x^2}{4} = \frac{2.56}{4}$ Divide both sides by 4 $x^2 = 0.64$	The equation $x^2 = 0.64$ represents _____.
$\sqrt{x^2} = \sqrt{0.64}$ Take the square root of both sides $x = 0.8$ and $x = -0.8$	The values $x = 0.8$ and $x = -0.8$ represent the tile's _____.
$x = 0.8$	One tile in the pattern has a length of 0.8 feet. A negative 0.8 feet does not make sense for this context.

2. How do the steps in the table relate to the steps you used in the Warm-Up to determine the measurements, in feet, for each tile?

3. Wyatt wondered if he could solve $4x^2 = 2.56$ by first taking the square root of both sides. His work is shown.

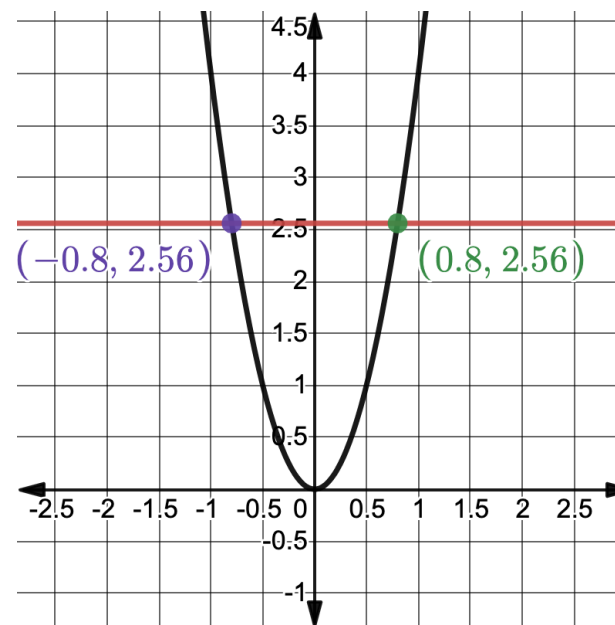
$$\sqrt{4x^2} = \sqrt{2.56}$$

$$2x = 1.6$$

$$x = 0.8$$

Wyatt's teacher told him that he could solve this quadratic by taking the square root of both sides first, but that he made a mistake in the second step of his work. Correct Wyatt's work.

4. Dominic wanted to find the solution to $4x^2 = 2.56$ using a graphing tool. He graphed $y = 4x^2$ and $y = 2.56$. His graph is shown.



Discuss with your partner how the graph shows the solutions to $4x^2 = 2.56$. Summarize your discussion.

5. Takeisha created a table of values for $y = 4x^2$.

x	-1	-0.9	-0.8	-0.7	-0.6	-0.5	-0.4	-0.3	-0.2	-0.1	0
y	4	3.24		1.96	1.44	1	0.64		0.16		0
x	1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	
y	4		2.56	1.96	1.44		0.64	0.36	0.16	0.04	

- Work with your partner to find the missing values in the table.
 - Describe the process you and your partner used to find the missing values.
 - How can Takeisha use the table of values to solve $4x^2 = 2.56$?
6. Out of all of these representations, the answer $4x^2 = 2.56$ that Aniyah should use is _____ ft because _____.



9.5.3 Guided Instruction: Solving Quadratics by Taking the Square Root

The square root of a number is the value that, when multiplied by itself, results in the radicand. For example, $\sqrt{25}$ is 5 or -5 .



The **principal square root** is the positive square root of a positive real number.

The principal square root is shown algebraically using $\sqrt{a^2} = |a|$ where $\{a|a \in R\}$. If $a < 0$ then $|a|$ is positive and if $a > 0$, $|a|$ is also positive.

In the Exploration, the steps to solving the equation $4x^2 = 2.56$ were shown.

The step $\sqrt{x^2} = \sqrt{0.64}$ resulted in $x = 0.8$ and $x = -0.8$. The steps below can be used to show this algebraically using the relationship $\sqrt{a^2} = |a|$ where $\{a|a \in R\}$.

$$\sqrt{x^2} = \sqrt{0.64}$$

$$|x| = 0.8$$

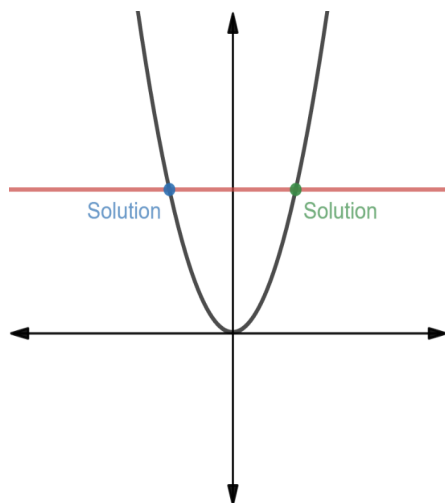
$$x = 0.8 \text{ and } x = -0.8$$

Because $\sqrt{a^2} = |a|$ for all real numbers, $\sqrt{x^2} = |x|$.

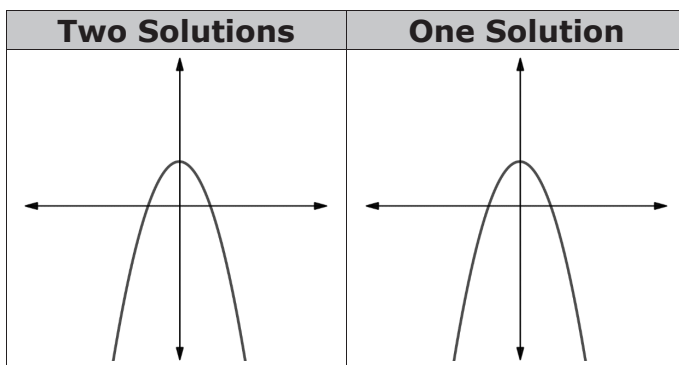
Since $|0.8| = 0.8$ and $|-0.8| = 0.8$ then the x in $|x|$ could be 0.8 or -0.8 .

Unless otherwise stated, the "square root" of a number refers **only** to the principal square root. When solving $x^2 = a$ using square roots, the answer should **always** be both the positive and negative values.

The graph demonstrates why there are two solutions when solving a quadratic.

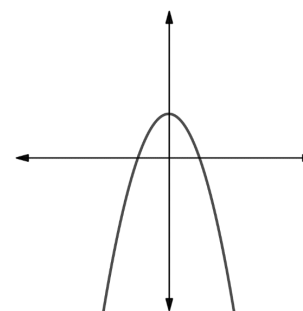


1. Work with your partner to answer the following questions.
 - a. Draw a horizontal line for $y = a$ to demonstrate the number of solutions.



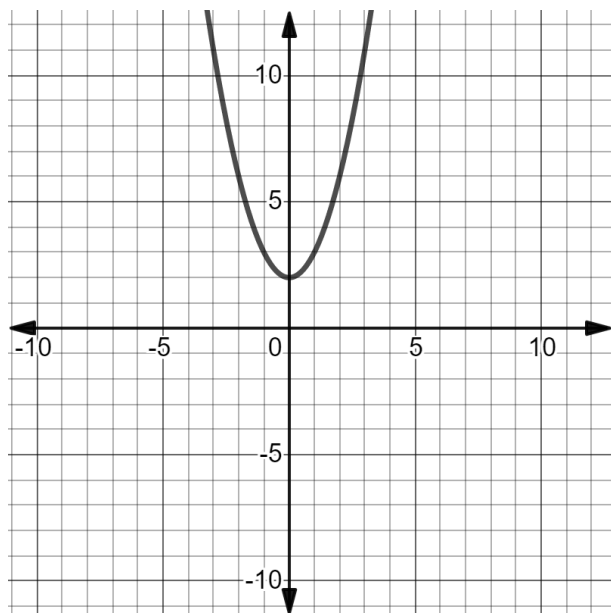
- b. Describe where the horizontal line intersects with the parabola when the quadratic has one solution.

- c. Discuss with your partner where a horizontal line could be drawn so that it does not intersect with the parabola. Demonstrate this by drawing a horizontal line.



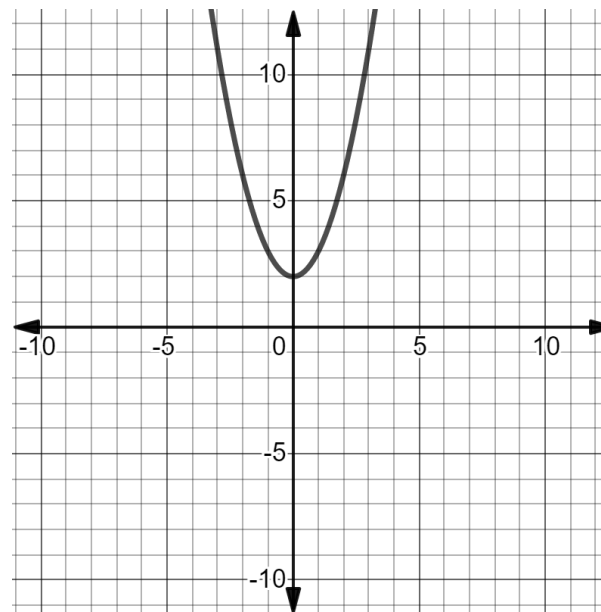
2. Consider the equation $x^2 + 2 = 2$.
 - a. Find the solution to $x^2 + 2 = 2$.

- b. The graph of $y = x^2 + 2$ is shown. Draw a horizontal line that shows $y = 2$.



- c. Explain how the location of the intersection of the horizontal line and the parabola shows that $x^2 + 2 = 2$ only has one solution.

3. The graph of $y = x^2 + 2$ is shown.



- a. Draw the horizontal line $y = -7$.
- b. Find the solution to $x^2 + 2 = -7$.

- When solving $x^2 = a$, where $\{a: a > 0\}$ the answer will always be both the positive and negative values of $\sqrt{a}$.
- When solving $x^2 = a$, where $\{a: a = 0\}$ the answer will be 0.
- When solving $x^2 = a$, where $\{a: a < 0\}$ the answer is non-real.

The topic of non-real numbers (imaginary numbers) will be learned in Algebra 2.



9.5.4 Guided Practice: Solving Quadratics by Taking the Square Root

Solve the equation. Show your work.

1. $x^2 - 41 = -5$



9.5.5 Try It: Solving Quadratics by Taking the Square Root

Solve each equation. Show your work.

1. $-5x^2 = -180$

2. $39 = 3(x^2 + 7)$



9.5.6 Wrap-Up: Ticket Out the Door

1. Find the solution(s). Show your work.

$$86 = 5x^2 - 34$$

Unit 9, Lesson 6: Solving Quadratic Equations by Completing the Square



Benchmarks

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can solve a quadratic by completing the square.
- ☐ I can interpret solutions to quadratics in a real-world context as viable or nonviable.